
SGIQ: Sparse-Gradient-Informed Gradient Estimation for Variational Quantum Circuits

Yujia Cai

Rochester Institute of Technology
Rochester, NY, USA
yc7054@rit.edu

Zhe Li

Rochester Institute of Technology
Rochester, NY, USA
zl4063@rit.edu

Chaowen Guan

University of Cincinnati
Cincinnati, OH, USA
guance@ucmail.uc.edu

Haibo Yang

Rochester Institute of Technology
Rochester, NY, USA
hbycis@rit.edu

Abstract

Gradient estimation is a major measurement bottleneck in training complex variational quantum circuits (VQCs) because backpropagation scaling is unattainable for general cases with unknown quantum inputs and no quantum memory. Existing gradient estimators therefore face a cost-accuracy trade-off: parameter-shift rule (PSR) incurs a cost linear in the parameter dimension, while simultaneous perturbation stochastic approximation (SPSA) uses few evaluations but can suffer from dimension-dependent variance. To improve this trade-off, we exploit settings in which loss-gradient energy concentrates on a small subset of parameters. We formalize this property through gradient-energy sparsity and derive sufficient conditions from causal-cone, commutation, and response-decay mechanisms. Building on this structure, we propose SGIQ, a **S**parse-**G**radient-**I**nformed **Q**uantum gradient estimation algorithm that uses averaged SPSA estimates to rank all circuit parameters, replaces the selected coordinate estimates with PSR gradients, and retains the SPSA estimates elsewhere. Its per-step cost scales with the circuit's gradient structure, rather than the parameter dimension. In ideal and finite-shot regression experiments, SGIQ uses fewer gradient circuits than PSR and attains lower normalized test mean absolute error (MAE) than SPSA baselines.

1 Introduction

Gradient estimation is a major measurement bottleneck in training variational quantum circuits (VQCs), whose measured outputs define trainable models for quantum machine learning and variational optimization [1, 2]. Unlike classical backpropagation, derivatives through a measured quantum circuit generally require additional quantum evaluations, and backpropagation scaling is unattainable for general quantum models with unknown quantum inputs and no quantum memory [3]. The parameter-shift rule (PSR) provides a standard approach to computing VQC gradients [4, 5], but its circuit cost scales linearly with the number of trainable parameters. For d trainable parameters, a mini-batch of B samples, and Q separately evaluated measurement groups per sample, full PSR requires $2BQd$ gradient-circuit evaluations per optimizer step. In contrast, simultaneous perturbation stochastic approximation (SPSA) [6] estimates all gradient coordinates from m paired perturbations using only $2BQm$ evaluations, independent of d when m is fixed. This reduction in cost, however, comes at the expense of estimation accuracy: each SPSA coordinate estimate mixes contributions from other coordinates, leading to dimension-dependent variance, while finite-shot measurements

introduce additional noise. More generally, reducing the per-step directional-gradient cost does not necessarily yield a proportional reduction in total training cost, since parameter dependence can reappear through estimator variance or convergence [3, 7]. Thus, PSR and SPSA expose a fundamental cost-accuracy trade-off in quantum gradient estimation.

This trade-off raises a natural question: *must all gradient coordinates be estimated with equal accuracy?* For structured VQCs, the answer can be no. Known causal-cone, commutation, and response-locality arguments show that some parameters can have zero or small derivatives [8–11]. These structures can cause the loss-gradient energy to concentrate on a small effective coordinate set of size $k \ll d$. In this regime, estimating every coordinate with equal precision is unnecessary, suggesting that gradient-evaluation resources should instead be allocated according to coordinate importance. We formalize this structure through *gradient-energy sparsity* and derive sufficient conditions under which exact or approximate gradient-energy concentration arises.

Building on this structure, we propose SG1Q, a Sparse-Gradient-Informed Quantum gradient estimation algorithm that combines the global coverage of SPSA with the coordinate-wise accuracy of PSR. SG1Q uses averaged SPSA estimates as a low-cost probe over all d circuit parameters and ranks the coordinates by their estimated gradient magnitudes. It then replaces the SPSA estimates on the selected top- k coordinates with PSR gradients, while retaining the SPSA estimates elsewhere. This allocation reduces the per-step gradient-circuit cost to a fraction $(m + k)/d$ of that of full PSR.

Existing measurement-efficient approaches explore different ways of reducing or allocating quantum gradient evaluations. Random coordinate descent evaluates randomly selected gradient coordinates [12], while Guided-SPSA combines SPSA and PSR across training samples [13]. Coyle et al. [7] develop a forward-gradient framework with adaptive measurement allocation. In contrast, SG1Q allocates accurate gradient evaluations across parameter coordinates according to their estimated importance while retaining an SPSA estimate over the full parameter space.

We summarize our contributions as follows. First, we formalize gradient-energy sparsity and derive sufficient conditions from known circuit mechanisms. Second, we introduce SG1Q, which selectively allocates accurate PSR evaluations to important gradient coordinates identified by SPSA. Third, we demonstrate gradient-energy concentration in structured VQC regression models and show, under both ideal and finite-shot simulation, that SG1Q reduces gradient-circuit cost relative to full PSR while outperforming SPSA.

2 Preliminaries

For circuit parameters $\theta = (\theta_1, \dots, \theta_d)$, define $\rho(\theta) = U(\theta)\rho_0 U^\dagger(\theta)$, where ρ_0 is the input density operator, $U(\theta)$ is the parameterized unitary, and $\rho(\theta)$ is the output density operator. For a Hermitian observable O , define the measured expectation value $E_O(\theta) = \text{Tr}[O\rho(\theta)]$. In supervised learning, measured outputs are combined through a classical readout and loss function to form the scalar mini-batch loss $L(\theta)$. The mini-batch and classical readout parameters are held fixed when estimating $g = \nabla_\theta L(\theta)$.

Assume that each parameter appears in a single Pauli-generated rotation $U_j(\theta_j) = e^{-i\theta_j P_j/2}$, with $P_j^2 = I$. For the j th standard basis vector $e_j \in \mathbb{R}^d$, PSR gives

$$\partial_j E_O(\theta) = \frac{1}{2} \left[E_O\left(\theta + \frac{\pi}{2} e_j\right) - E_O\left(\theta - \frac{\pi}{2} e_j\right) \right].$$

This identity is exact for exact expectation values [4]. Finite-shot evaluations add sampling error, and extensions to broader generators are available [5]. In supervised learning, PSR differentiates the measured outputs, and the chain rule yields the loss gradient. SPSA perturbs all parameters simultaneously [6]. For m independent Rademacher directions $\Delta^{(r)} \in \{-1, +1\}^d$ and perturbation radius $c > 0$, define

$$h_j^{(r)} = \frac{E_O(\theta + c\Delta^{(r)}) - E_O(\theta - c\Delta^{(r)})}{2c\Delta_j^{(r)}}, \quad \frac{1}{m} \sum_{r=1}^m h_j^{(r)} \approx \partial_j E_O(\theta). \quad (1)$$

Thus one pair of circuit evaluations estimates all d partial derivatives at once, but each directional difference mixes contributions from all coordinates, and finite-shot evaluations add measurement noise.

3 Gradient-energy sparsity in VQCs

3.1 Gradient-energy sparsity

Let $L : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable scalar loss and $g(\boldsymbol{\theta}) = \nabla L(\boldsymbol{\theta})$. For $S \subseteq [d] = \{1, \dots, d\}$, let $\mathbf{g}_S \in \mathbb{R}^d$ agree with $\mathbf{g}$ on S and vanish elsewhere, so that $\|\mathbf{g}_S\|_2^2 = \sum_{j \in S} g_j^2$.

Definition 1 (Gradient-energy sparsity). *For $k \in [d]$ and $\rho \in (0, 1]$, $g(\boldsymbol{\theta})$ is (k, ρ) -sparse if a set $S \subseteq [d]$ with $|S| = k$ satisfies*

$$\|\mathbf{g}_S(\boldsymbol{\theta})\|_2^2 \geq \rho \|\mathbf{g}(\boldsymbol{\theta})\|_2^2. \quad (2)$$

When $\rho = 1$, $\mathbf{g}$ is supported on at most k coordinates; when ρ is close to one and $k \ll d$, its energy is concentrated on k coordinates up to a small tail. Unlike exact sparsity, this criterion is stable under small perturbations of $\mathbf{g}$, which suits finite-shot and stochastic optimization.

For a nonzero gradient, let $|g|_{(1)} \geq \dots \geq |g|_{(d)}$ denote its ordered coordinate magnitudes and define $k_\rho(\boldsymbol{\theta}) := \min\{k : \sum_{j=1}^k |g|_{(j)}^2 \geq \rho \|\mathbf{g}(\boldsymbol{\theta})\|_2^2\}$.

3.2 Circuit conditions for gradient-energy sparsity

We give circuit conditions under which the observable gradient $\nabla_{\boldsymbol{\theta}} E_O$ satisfies Definition 1. $U(\boldsymbol{\theta}) = U_{>j} U_j(\theta_j) U_{<j}$ with $U_j(\theta_j) = e^{-i\theta_j G_j/2}$ and $\|G_j\| \leq 1$, where $U_{<j}$ and $U_{>j}$ contain the gates before and after U_j . With the observable $O_{>j} = U_{>j}^\dagger O U_{>j}$ and the state $\rho_j = U_j U_{<j} \rho_0 U_{<j}^\dagger U_j^\dagger$, direct differentiation gives

$$\partial_j E_O(\boldsymbol{\theta}) = \frac{i}{2} \text{Tr}(\rho_j [G_j, O_{>j}]), \quad [A, B] := AB - BA. \quad (3)$$

Throughout, write $O = \sum_{\alpha} a_{\alpha} O_{\alpha}$ with real coefficients a_{α} and Hermitian terms O_{α} supported on small qubit sets X_{α} .

Backward causal cones. The backward causal cone of O_{α} is obtained by starting from X_{α} at the output and, moving toward the input, collecting every gate that acts on a qubit already in the set and adding that gate's qubits to the set; gates outside the cone cancel in $U^\dagger O_{\alpha} U$. Let $\mathcal{C}_O$ contain the parameter indices whose gates lie in the cone of at least one O_{α} .

Proposition 1 (Causal-cone sparsity). *If $j \notin \mathcal{C}_O$, then $\partial_j E_O(\boldsymbol{\theta}) = 0$ for all $\boldsymbol{\theta}$. Hence, if $|\mathcal{C}_O| \leq k$, $\nabla_{\boldsymbol{\theta}} E_O$ is $(k, 1)$ -sparse.*

For shallow, geometrically local circuits with local readouts, the cone can contain far fewer parameterized gates than the full ansatz [8, 9]. The proof and a five-qubit example are in Appendix A.1.

Commutation. Inside the cone, a derivative still vanishes when the generator commutes with $O_{>j}$.

Proposition 2 (Commutation-induced sparsity). *Let $\mathcal{A}_O(\boldsymbol{\theta}) = \{j : [G_j, O_{>j}(\boldsymbol{\theta})] \neq 0\}$. If $j \notin \mathcal{A}_O(\boldsymbol{\theta})$, then $\partial_j E_O(\boldsymbol{\theta}) = 0$. Hence, if $|\mathcal{A}_O(\boldsymbol{\theta})| \leq k$, $\nabla_{\boldsymbol{\theta}} E_O(\boldsymbol{\theta})$ is $(k, 1)$ -sparse.*

This follows directly from Eq. 3. Related commutator and Lie-algebraic representations appear in analytic quantum-gradient methods [4, 5, 10]. The proof and a diagonal Ising example are provided in Appendix A.2.

Decaying response tails. Exact zeros are not necessary for concentration, it suffices that the response $|\partial_j E_{O_{\alpha}}|$ of a term decays with its distance from the gate. Let $\text{dist}(X_{\alpha}, \text{supp } G_j)$ be the shortest-path distance between the two qubit sets in the interaction graph, whose edges join qubits coupled by a two-qubit gate.

Assumption 1 (Decaying response envelope). *There exist $L_h > 0$ and a nonincreasing $\chi : [0, \infty) \rightarrow [0, 1]$ with $\lim_{r \rightarrow \infty} \chi(r) = 0$ such that $|\partial_j E_{O_{\alpha}}(\boldsymbol{\theta})| \leq L_h \chi(\text{dist}(X_{\alpha}, \text{supp } G_j))$ for all $\boldsymbol{\theta}$, inputs j , and α .*

By Eq. 3, $|\partial_j E_{O_{\alpha}}| \leq \frac{1}{2} \| [G_j, U_{>j}^\dagger O_{\alpha} U_{>j}] \|_{\text{op}}$, so Lieb–Robinson-type bounds yield such envelopes, e.g., for power-law interactions [11]. For $\tau > 0$, let $A_j = \sum_{\alpha} |a_{\alpha}| \chi(\text{dist}(X_{\alpha}, \text{supp } G_j))$ and $\mathcal{T}_{\tau}(O) = \{j : A_j > \tau\}$ with $k = |\mathcal{T}_{\tau}(O)| \geq 1$.

Algorithm 1 SGIQ for one mini-batch

Require: Mini-batch $\mathcal{B}$, parameters θ , directions m , selected coordinates k , radius c .

- 1: Forward pass: evaluate $z_{iq} = E_{O_q}(\theta; \mathbf{x}_i)$ and compute λ_{iq} by Eq. 4.
 - 2: For each circuit (i, q) and $r = 1, \dots, m$, draw $\Delta^{(i,q,r)} \sim \text{Unif}(\{-1, +1\}^d)$ and evaluate $E_{O_q}(\theta \pm c\Delta^{(i,q,r)}; \mathbf{x}_i)$.
 - 3: Form $\bar{g}$ from Eqs. 1 and 4.
 - 4: $\mathcal{S} \leftarrow \text{TopK}(|\bar{g}|, k)$.
 - 5: For $j \in \mathcal{S}$, evaluate $E_{O_q}(\theta \pm \frac{\pi}{2} \mathbf{e}_j; \mathbf{x}_i)$ and form g_j^{PSR} .
 - 6: $\tilde{g}_j \leftarrow g_j^{\text{PSR}}$ for $j \in \mathcal{S}$ and $\tilde{g}_j \leftarrow \bar{g}_j$ otherwise.
 - 7: **return** $\tilde{g}$ to the optimizer.
-

Proposition 3 (Compressibility from decaying responses). *For every $j \notin \mathcal{T}_\tau(O)$, $|\partial_j E_O(\theta)| \leq L_h \tau$, so $\|(\nabla E_O)_{\mathcal{T}_\tau^c}\|_2^2 \leq (d - k)L_h^2 \tau^2$. Consequently, for any $\rho \in (0, 1)$, $\nabla E_O(\theta)$ is (k, ρ) -sparse whenever $\|(\nabla E_O)_{\mathcal{T}_\tau}\|_2^2 \geq \frac{\rho}{1-\rho}(d - k)L_h^2 \tau^2$.*

Unlike the first two mechanisms, this one yields approximate rather than exact concentration; the proof is in Appendix A.3.

3.3 From circuit structure to loss gradients

The mechanisms above act on circuit outputs, whereas training uses the loss gradient. For sample $\mathbf{x}_i$ with target $\tilde{y}_i$ and observable O_q , let $z_{iq}(\theta) = E_{O_q}(\theta; \mathbf{x}_i)$, $\hat{y}_i = F_\phi(z_{i1}, \dots, z_{iQ})$ with readout parameters ϕ , and $\mathcal{L}_\mathcal{B}(\theta, \phi) = |\mathcal{B}|^{-1} \sum_{i \in \mathcal{B}} \ell(\hat{y}_i, \tilde{y}_i)$. The chain rule gives

$$\partial_j \mathcal{L}_\mathcal{B} = \sum_{i \in \mathcal{B}} \sum_{q=1}^Q \lambda_{iq} \partial_j E_{O_q}(\theta; \mathbf{x}_i), \quad \lambda_{iq} := \frac{1}{|\mathcal{B}|} \frac{\partial \ell(\hat{y}_i, \tilde{y}_i)}{\partial \hat{y}_i} \frac{\partial F_\phi}{\partial z_{iq}}. \quad (4)$$

Hence a loss-gradient coordinate vanishes if it vanishes for every observable and sample, and is small when these derivatives are small and the weights λ_{iq} are bounded; since the dominant coordinates may vary across samples, observables, and iterates, Section 5 measures concentration directly on the mini-batch loss gradient.

4 SGIQ

4.1 Gradient estimator

At each step, every circuit (sample $\mathbf{x}_i$, measurement group q) is differenced along its own m independent Rademacher directions by Eq. 1, and the chain rule Eq. 4, with λ_{iq} from the forward pass, combines the results into the averaged SPSA loss-gradient estimate $\bar{g}$. Each coordinate receives the score $s_j = |\bar{g}_j|$, and $\mathcal{S} = \text{TopK}(s, k)$ collects the indices of the k largest scores, with ties broken by a fixed rule. For $j \in \mathcal{S}$, PSR differentiates the circuit outputs and the chain rule gives g_j^{PSR} . The estimator is

$$\tilde{g}_j = \begin{cases} g_j^{\text{PSR}}, & j \in \mathcal{S}, \\ \bar{g}_j, & j \notin \mathcal{S}. \end{cases} \quad (5)$$

4.2 Theoretical guarantees

Setting. In the *ideal* setting circuit outputs are exact expectation values, in the *finite-shot* setting each is estimated from N_{sh} shots. With θ and $\mathcal{B}$ fixed, $\mathbb{E}[\cdot]$ averages over the SPSA directions and, under finite shots, the measurement outcomes. The target is $\mathbf{g} := \nabla_\theta \mathcal{L}_\mathcal{B}(\theta)$, and $\nabla \mathcal{J}$ is the gradient of $\mathcal{J} := \mathbb{E}_\mathcal{B}[\mathcal{L}_\mathcal{B}]$. We write $Q = 1$ and $\lambda_i := \lambda_{i1}$. By Eq. 4, $g_j = \sum_{i \in \mathcal{B}} \Gamma_{ij}$ with $\Gamma_{ij} := \lambda_i \partial_j E_O(\theta; \mathbf{x}_i)$. Since each circuit has its own directions, the estimator noise is governed by this *per-sample gradient matrix* $\Gamma \in \mathbb{R}^{B \times d}$, whose energy is $\|\Gamma\|_F^2 = \sum_i \lambda_i^2 \|\nabla_\theta E_O(\theta; \mathbf{x}_i)\|_2^2$. The Taylor expansion of the SPSA difference quotient (Appendix B.1) shows that coordinate j of $\mathbf{g}$ has

bias $c^2 \bar{\beta}_j + O(c^4)$, with

$$\bar{\beta}_j := \sum_{i \in \mathcal{B}} \lambda_i b_{ij}, \quad b_{ij} := \frac{1}{6} \left[3 \sum_{p=1}^d \partial_j \partial_p^2 E_O - 2 \partial_j^3 E_O \right] (\boldsymbol{\theta}; \mathbf{x}_i). \quad (6)$$

Under finite shots, let $\sigma_{\mathcal{P},j}^2$ and $\sigma_{\mathcal{S}}^2$ be the shot variances of a PSR coordinate and of an SPSA coordinate from one direction, and $\sigma_{\mathcal{P}}^2 := \max_j \sigma_{\mathcal{P},j}^2$. Since a measured output has variance at most $\|O\|_{\text{op}}^2 / N_{\text{sh}}$,

$$\sigma_{\mathcal{P}}^2 \leq s / N_{\text{sh}}, \quad \sigma_{\mathcal{S}}^2 \leq s / (N_{\text{sh}} c^2), \quad s := \frac{1}{2} \|O\|_{\text{op}}^2 \sum_{i \in \mathcal{B}} \lambda_i^2. \quad (7)$$

In the ideal setting all shot terms vanish and we set $s := 0$. The *realized concentration* of the selected set is $\hat{\rho} := \frac{\|\mathbf{g}_{\mathcal{S}}\|_2^2}{\|\mathbf{g}\|_2^2} \in [0, 1]$, so that $\mathbf{g}$ is $(k, \hat{\rho})$ -sparse in the sense of Definition 1. For a random vector $\mathbf{v}$, we write $\text{Var}(\mathbf{v}) := \mathbb{E}\|\mathbf{v} - \mathbb{E}\mathbf{v}\|_2^2$ for its total variance, the sum of its coordinate variances.

Assumption 2 (Predictable selection). *$\mathcal{S}$ is independent of the directions and shots used to form $\bar{\mathbf{g}}$ in the current step.*

Algorithm 1 selects $\mathcal{S}$ from the same $\bar{\mathbf{g}}$ and violates Assumption 2; Lemma 3 shows that, under a gap condition, it coincides with a predictable estimator except with probability exponentially small in m .

Lemma 1 (Bias). *Under Assumption 2, in both settings, $\mathbb{E}[\tilde{g}_j] = g_j$ for $j \in \mathcal{S}$ and $\mathbb{E}[\tilde{g}_j] - g_j = c^2 \bar{\beta}_j + O(c^4)$ for $j \notin \mathcal{S}$, with $\bar{\beta}_j$ from Eq. 6. Hence*

$$F(k) := \|\mathbb{E}[\tilde{\mathbf{g}}] - \mathbf{g}\|_2^2 = \sum_{j \notin \mathcal{S}} (c^2 \bar{\beta}_j)^2 + O((d-k)c^6) = (d-k) c^4 \bar{\beta}_{\mathcal{S}^c}^2 + O((d-k)c^6), \quad (8)$$

where $\bar{\beta}_{\mathcal{S}^c}^2 := \frac{1}{d-k} \sum_{j \notin \mathcal{S}} \bar{\beta}_j^2$ is the mean squared bias coefficient over the unselected coordinates.

The bias does not depend on m and vanishes at $k = d$. $F(0)$ is the bias of averaged SPSA. Because the difference quotient is formed on circuit outputs before chaining, shot noise leaves it unchanged (Appendix B.6).

Lemma 2 (Variance). *Under Assumption 2, conditionally on $\mathcal{S}$,*

$$\text{Var}(\tilde{\mathbf{g}}) = \frac{(d-k) \|\Gamma\|_F^2 - \|\Gamma_{\mathcal{S}^c}\|_F^2}{m} + \frac{(d-k) \sigma_{\mathcal{S}}^2}{m} + \sum_{j \in \mathcal{S}} \sigma_{\mathcal{P},j}^2 + O\left(\frac{(d-k)c^2}{m}\right), \quad (9)$$

where $\Gamma_{\mathcal{S}^c}$ keeps the unselected columns of Γ .

Proposition 4 (Gradient estimation error). (i) *For any selection rule,*

$$\mathbb{E}\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 = \mathbb{E}\|\bar{\mathbf{g}} - \mathbf{g}\|_2^2 - \mathbb{E}\left[\sum_{j \in \mathcal{S}} ((\bar{g}_j - g_j)^2 - \sigma_{\mathcal{P},j}^2)\right]. \quad (10)$$

In the ideal setting $\sigma_{\mathcal{P},j} = 0$, so $\mathbb{E}\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 \leq \mathbb{E}\|\bar{\mathbf{g}} - \mathbf{g}\|_2^2$; under finite shots and Assumption 2, the same holds when $\sigma_{\mathcal{P}}^2 \leq \sigma_{\mathcal{S}}^2/m$. (ii) Under Assumption 2, $\mathbb{E}\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 = F(k) + \text{Var}(\tilde{\mathbf{g}})$.

Proposition 5 (Descent-direction reliability). *Under Assumption 2, $\mathbb{E}\langle \mathbf{g}, \tilde{\mathbf{g}} - \mathbf{g} \rangle = c^2 \langle \mathbf{g}_{\mathcal{S}^c}, \bar{\beta}_{\mathcal{S}^c} \rangle + O(c^4)$ and*

$$\text{Var}(\langle \mathbf{g}, \tilde{\mathbf{g}} \rangle) \leq \frac{(1-\hat{\rho}) \|\mathbf{g}\|_2^2}{m} (2\|\Gamma\|_F^2 + \sigma_{\mathcal{S}}^2) + \hat{\rho} \|\mathbf{g}\|_2^2 \sigma_{\mathcal{P}}^2 + O\left(\frac{c^2}{m}\right). \quad (11)$$

In the ideal setting, averaged SPSA is the case $\hat{\rho} = 0$: the directional-variance bound of SG1Q is that of SPSA multiplied by $1 - \hat{\rho}$, so the more concentrated the gradient, the more reliable the descent direction.

Selection by the SPSA scores. Let $\mathcal{S}_*$ index the k largest $|g_j|$, let $\rho_k := \|\mathbf{g}_{\mathcal{S}_*}\|_2^2 / \|\mathbf{g}\|_2^2$ be the true top- k coverage (reported in Table 1 for $k = k_\rho$), let $b_\infty := \max_j |\mathbb{E}\tilde{g}_j - g_j| = O(c^2)$, and let $\gamma := |g|_{(k)} - |g|_{(k+1)}$.

Lemma 3 (Selection consistency and fidelity). *For the in-sample rule $\mathcal{S} = \text{TopK}(|\bar{g}|, k)$ of Algorithm 1 and $\epsilon := \max_j |\bar{g}_j - g_j|$,*

$$\hat{\rho} \leq \rho_k, \quad \sqrt{\hat{\rho}} \geq \sqrt{\rho_k} - \frac{2\sqrt{k}\epsilon}{\|\mathbf{g}\|_2}. \quad (12)$$

If $\gamma > 2b_\infty$, then $\mathcal{S} = \mathcal{S}_$, and hence $\hat{\rho} = \rho_k$, with probability at least $1 - p_{\text{fail}}$, where*

$$p_{\text{fail}} \leq 2d \exp\left(-\frac{m t_\gamma^2}{2\sigma_X^2 + \frac{2}{3}R t_\gamma}\right), \quad t_\gamma := \frac{\gamma}{2} - b_\infty, \quad (13)$$

so that $(1 - p_{\text{fail}}) \rho_k \leq \mathbb{E}[\hat{\rho}] \leq \rho_k$. Here $\sigma_X^2 = \|\Gamma\|_F^2 + \sigma_S^2 + O(c^2)$ and R bound the variance and the range of the contribution of a single direction to $\bar{g}_j$.

Outside an event of probability p_{fail} , which is small once the SPSA noise $\sigma_X/\sqrt{m}$ is well below the gap, Algorithm 1 returns the estimator built on the fixed set $\mathcal{S}_*$; this estimator satisfies Assumption 2, so the results above apply with $\hat{\rho} = \rho_k$. Without a gap, Eq. 12 still ties $\hat{\rho}$ to ρ_k (Appendix B.9).

We analyze SGD on θ with the readout fixed, $\theta_{t+1} = \theta_t - \eta \tilde{\mathbf{g}}_t$, where $\tilde{\mathbf{g}}_t$ is formed on an independently drawn mini-batch $\mathcal{B}_t$ and $\mathcal{S}_t$ is chosen at step t . The experiments use Adam; the estimator results above do not depend on the optimizer.

Assumption 3 (Objective and sampling). *$\mathcal{J}$ is bounded below by $\mathcal{J}_*$, and $\nabla \mathcal{J}$ is L_g -Lipschitz. Mini-batches satisfy $\mathbb{E}[\mathbf{g}_{\mathcal{B}_t} | \theta_t] = \nabla \mathcal{J}(\theta_t)$ and $\mathbb{E}[\|\mathbf{g}_{\mathcal{B}_t} - \nabla \mathcal{J}(\theta_t)\|_2^2 | \theta_t] \leq \sigma_{\mathcal{B}}^2$, and $\bar{G} := \sup_t \|\Gamma_t\|_F^2$, $\bar{s} := \sup_t s_t$, and $B_\beta^2 := \sup_t \bar{\beta}^2_{\mathcal{S}_t}$ are finite, where Γ_t , s_t (Eq. 7) and $\bar{\beta}^2_{\mathcal{S}_t}$ (Lemma 1) are taken at step t .*

Theorem 1 (Convergence of SGiQ). *Under Assumption 3, let $0 < \eta \leq 1/(4L_g)$ and $\Delta_{\mathcal{J}} := \mathcal{J}(\theta_0) - \mathcal{J}_*$. (i) If every $\mathcal{S}_t$ satisfies Assumption 2 given the history,*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla \mathcal{J}(\theta_t)\|_2^2 \leq \frac{2\Delta_{\mathcal{J}}}{\eta T} + L_g \eta v(k) + 3F(k), \quad (14)$$

where $F(k) \leq (d-k)c^4 B_\beta^2 + \mathcal{O}((d-k)c^6)$ bounds the bias of Lemma 1 along the run, $v(k) := V(k) + (\sigma_{\mathcal{B}} + \sqrt{F(k)})^2$, and, by Lemma 2 and Eq. 7,

$$V(k) = \frac{d-k}{m} \left(\bar{G} + \frac{\bar{s}}{N_{\text{sh}} c^2} \right) + \frac{k \bar{s}}{N_{\text{sh}}} + \mathcal{O}\left(\frac{(d-k)c^2}{m}\right). \quad (15)$$

(ii) For any selection rule, including Algorithm 1, Eq. 14 holds with $F(k)$ and $v(k)$ replaced by e^2 and $(e + \sigma_{\mathcal{B}})^2$, where $e^2 := \sup_t \mathbb{E}_t \|\tilde{\mathbf{g}}_t - \mathbf{g}_{\mathcal{B}_t}\|_2^2$ is bounded by Proposition 4(i).

Unlike the variance term, the bias term $3F(k)$ is not controlled by η . If the gap of Lemma 3 holds at every step, Algorithm 1 coincides with the predictable variant using $\mathcal{S}_{*,t}$ with probability at least $1 - Tp_{\text{fail}}$. Since $G_\infty := \sup_\theta \|\nabla \mathcal{J}(\theta)\|_2^2 < \infty$ for bounded readout weights, the bounds of part (i) and Corollary 1 then hold for Algorithm 1 itself up to an additive $Tp_{\text{fail}}G_\infty$, which is negligible once m grows logarithmically in dT (Appendix B.9).

Let $A_0 := 8L_g \Delta_{\mathcal{J}} + \sigma_{\mathcal{B}}^2/2$ denote the part shared by all estimators.

Corollary 1 (Explicit rates). *Let $\eta = 1/(4L_g \sqrt{T})$. Under Theorem 1(i), up to terms of order $(d-k)c^2/(m\sqrt{T}) + (d-k)c^6$,*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla \mathcal{J}(\theta_t)\|_2^2 \leq \underbrace{\frac{1}{\sqrt{T}} \left[A_0 + \frac{d-k}{4m} \left(\bar{G} + \frac{\bar{s}}{N_{\text{sh}} c^2} \right) + \frac{k \bar{s}}{4N_{\text{sh}}} \right]}_{\text{convergence rate}} + \underbrace{\frac{7}{2}(d-k)c^4 B_\beta^2}_{\text{accuracy floor}}. \quad (16)$$

(a) Ideal setting ($\bar{s} = 0$). If $c \leq (d-k)^{-1/4} T^{-1/8}$ (no condition when $k = d$), then

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla \mathcal{J}(\theta_t)\|_2^2 \leq \frac{K(m, k, d)}{\sqrt{T}} + \mathcal{O}(T^{-3/4}), \quad (17)$$

where $K(m, k, d) := A_0 + \frac{7}{2}B_\beta^2 + (d - k)\bar{G}/(4m)$, with $B_\beta := 0$ at $k = d$. (b) Finite-shot setting. For a fixed c , the bound converges at rate $\mathcal{O}(T^{-1/2})$ to the accuracy floor. With $c = c_\star := (\bar{s}/(28mN_{\text{sh}}B_\beta^2\sqrt{T}))^{1/6}$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla \mathcal{J}(\theta_t)\|_2^2 \leq \frac{K_{\text{sh}}}{\sqrt{T}} + \frac{C_{\text{sh}}}{T^{1/3}}, \quad (18)$$

where $K_{\text{sh}} := A_0 + (d - k)\bar{G}/(4m) + k\bar{s}/(4N_{\text{sh}})$ and $C_{\text{sh}} := 1.2(d - k)(B_\beta\bar{s}/(mN_{\text{sh}}))^{2/3}$.

Eq. 16 separates the convergence rate, whose algorithm-dependent part scales as $(d - k)/m$, from the accuracy floor, which depends on neither m nor T and vanishes at $k = d$. In the ideal setting, K interpolates between averaged SPSA ($k = 0$) and PSR ($K = A_0$ at $k = d$).

Corollary 2 (Equal-budget comparison). *Let SPSA use $m' = m + k$ directions, so that both methods evaluate $2BQ(m + k)$ gradient circuits per step, and let Assumption 2 hold. (i) Bias: $F_{\text{SPSA}} - F_{\text{SGiQ}} = c^4\|\beta_S\|_2^2 + \mathcal{O}(c^6) \geq 0$, independent of m . (ii) Descent direction: in the ideal setting and up to $\mathcal{O}(c^2)$ terms, the bound of Proposition 5 is smaller than its SPSA counterpart $2\|g\|_2^2\|\Gamma\|_F^2/m'$ if and only if $\hat{\rho} > k/(m + k)$.*

4.3 Gradient-circuit cost

Let $B = |\mathcal{B}|$ and let Q be the number of separately evaluated measurement groups per sample. PSR evaluates two shifted circuits per parameter, SPSA two per direction, and SGiQ additionally two per selected coordinate, so the per-step costs are $N_{\text{PSR}} = 2BQd$, $N_{\text{SPSA}} = 2BQm$, and $N_{\text{SGiQ}} = 2BQ(m + k)$, with $m + k \leq d$.

Corollary 3 (Total cost). *Over T steps, method X uses $C_X = TN_X$ gradient circuits and $N_{\text{sh}}C_X$ shots, i.e., $C_{\text{PSR}} = 2BQdT$, $C_{\text{SPSA}} = 2BQm'T$ with m' directions, and $C_{\text{SGiQ}} = 2BQ(m + k)T$. In the ideal setting of Corollary 1(a), the average in Eq. 17 is at most ε , up to its $\mathcal{O}(T^{-3/4})$ term, after $T(\varepsilon) = K(m, k, d)^2/\varepsilon^2$ steps, which requires*

$$C_{\text{SGiQ}}(\varepsilon) = \frac{2BQ(m + k)K(m, k, d)^2}{\varepsilon^2}, \quad C_{\text{PSR}}(\varepsilon) = \frac{2BQdA_0^2}{\varepsilon^2} \quad (19)$$

gradient circuits. Under finite shots with $c = c_\star$ (Corollary 1(b)), $2BQ(m + k)T(\varepsilon)$ circuits suffice, with $T(\varepsilon) = \max\{(2K_{\text{sh}}/\varepsilon)^2, (2C_{\text{sh}}/\varepsilon)^3\}$.

To reach accuracy ε , the dimension enters $C_{\text{SGiQ}}(\varepsilon)$ explicitly only through $m + k$ and $(d - k)/m$, and the guarantee of SGiQ needs fewer circuits than that of PSR whenever $(m + k)K(m, k, d)^2 < dA_0^2$.

5 Experiments

We first measure loss-gradient energy concentration in two structured VQC regression models and then compare SGiQ with full PSR and SPSA under ideal and finite-shot simulation.

5.1 Experimental setup and circuits

Datasets. We use the Friedman1 and Friedman2 regression benchmarks [14] with Gaussian target noise of standard deviation 0.1; Friedman1 has five input features and Friedman2 has four. Each feature is min-max scaled to $[0, \pi]$ and the target to $[-1, 1]$, and all errors are reported on the normalized target scale. The Friedman2 experiments use 100 training, 20 validation, and 80 test examples, with normalization computed on the full sample pool. The Friedman1 protocol is given in Section 5.3.

Models & Training Objective. The Friedman2 model (Fig. 1) has eight qubits, three variational layers, and $d = 48$ circuit parameters; it measures eight local observables and uses the affine readout $\hat{y}_i = b + \sum_{q=0}^7 w_q \langle Z_q \rangle_{\theta, \tilde{x}_i}$. The Friedman1 model (Fig. 2) uses the layered R_y - R_z and CZ-ring pattern of hardware-efficient circuits [15], with five qubits, eight layers, $d = 80$, and the readout $\hat{y}_i = a \langle Z^{\otimes 5} \rangle_{\theta, \tilde{x}_i}$ with a trainable scale a . Both models minimize the mini-batch mean squared error

(MSE) $\mathcal{L}_{\mathcal{B}} = |\mathcal{B}|^{-1} \sum_{i \in \mathcal{B}} (\hat{y}_i - \tilde{y}_i)^2$ without regularization. The quantum-gradient estimators act on θ , and the readout parameters are differentiated automatically; every estimator differentiates circuit outputs and applies the chain rule of Eq. 4.

Both circuits instantiate the commutation mechanism of Section 3: the final-layer R_z parameters ($\theta_{40}, \dots, \theta_{47}$ in Friedman2 and $\theta_{75}, \dots, \theta_{79}$ in Friedman1) are followed only by CZ gates and Z -type measurements, which commute with their generators, so these parameters have identically zero loss gradients.

Optimization & Model Selection. Circuit parameters use learning rate 0.03 and readout parameters 0.1. SPSA and SG1Q use $c = 0.1$; SPSA- m denotes SPSA with m directions, and SG1Q (m, k) uses m directions and k selected coordinates. After each epoch we evaluate the validation mean absolute error, $\text{MAE} = n^{-1} \sum_{i=1}^n |\hat{y}_i - \tilde{y}_i|$ over the n examples of a split, and report the test MAE and MSE of the checkpoint with the lowest validation MAE. The Friedman2 runs use Adam with $(\beta_1, \beta_2) = (0.5, 0.9)$.

Simulation. The ideal experiments use statevector expectation values and have no shot noise. The finite-shot experiment uses a local Aer simulator with 1024 shots per circuit evaluation and no gate, readout, depolarizing, or device-calibrated noise model.

For Friedman1, $Q = 1$.

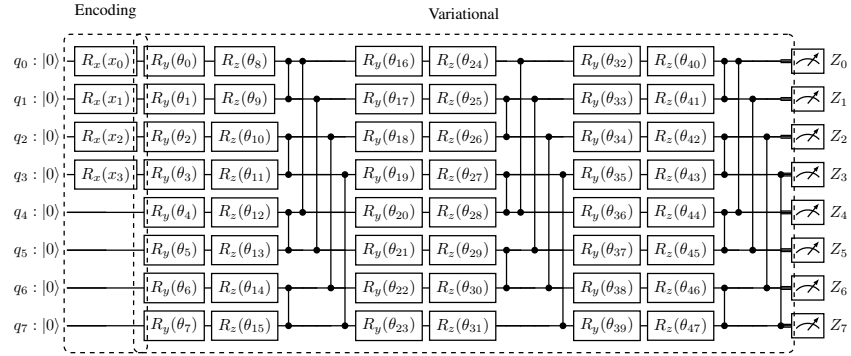


Figure 1: Friedman2 circuit ($d = 48$). q_0 – q_3 encode the features with $R_x(x_j)$; each of three R_y – R_z layers is followed by nearest-neighbor and feature-pair CZ gates $(0, 4), \dots, (3, 7)$. The readout measures $Z_0, \dots, Z_7$.

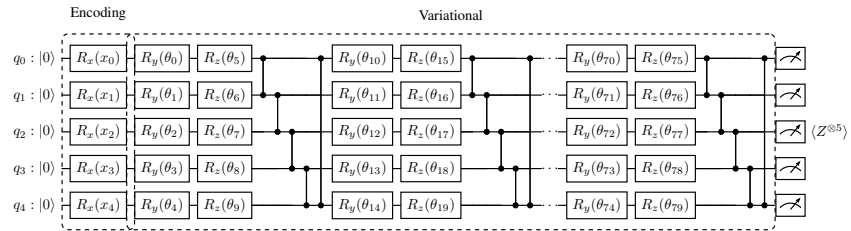


Figure 2: Friedman1 circuit ($d = 80$). Five features are encoded with $R_x(x_j)$; each of eight R_y – R_z layers is followed by the CZ ring $(0, 1), \dots, (4, 0)$, and layers 3–7 are elided. The readout is $\langle Z^{\otimes 5} \rangle$.

5.2 Gradient concentration in structured VQCs

We measure the gradient of the complete mini-batch MSE with respect to the quantum parameters. At the beginning of each run, one training mini-batch of size 20 is fixed as a probe batch. At epochs 5, 10, 15, and 20, full PSR computes $\mathbf{g} = \nabla_{\theta} \mathcal{L}_{\mathcal{B}_{\text{probe}}}(\theta, \phi)$.

Table 1 reports, for this $\mathbf{g}$, the count k_{ρ} defined in Section 3.1 and its coverage $\sum_{j=1}^{k_{\rho}} |g|_{(j)}^2 / \|\mathbf{g}\|_2^2$; k_{ρ} is a post-hoc diagnostic and is distinct from the manually chosen algorithmic budget k .

Table 1: PSR loss-gradient energy concentration on a fixed training probe batch. k_ρ is the minimum number of coordinates capturing ρ of the total squared-gradient magnitude.

Epoch	Friedman2 ($d = 48, \rho = 0.90$)			Friedman1 ($d = 80, \rho = 0.80$)		
	k_ρ	k_ρ/d	Coverage	k_ρ	k_ρ/d	Coverage
5	9	0.1875	0.9219	25	0.3125	0.8078
10	12	0.2500	0.9139	29	0.3625	0.8069
15	5	0.1042	0.9067	22	0.2750	0.8030
20	6	0.1250	0.9133	26	0.3250	0.8082

Table 2: Normalized test errors and gradient-circuit evaluations per training run.

Setting	Method	Test MAE	Test MSE	Cost
Ideal	PSR	0.08489 ± 0.01192	0.01304 ± 0.00261	960,000
	SPSA-30	0.10924 ± 0.01759	0.02122 ± 0.00370	360,000
	SPSA-60	0.10454 ± 0.01799	0.01888 ± 0.00601	720,000
	SGIQ (40, 10)	0.09504 ± 0.00342	0.01647 ± 0.00197	600,000
Finite-shot (1024 shots)	PSR	0.10865 ± 0.02196	0.02094 ± 0.01101	960,000
	SPSA-20	0.11579 ± 0.01090	0.02265 ± 0.00449	240,000
	SPSA-60	0.12070 ± 0.00264	0.02559 ± 0.00317	720,000
	SGIQ (40, 20)	0.10714 ± 0.01119	0.02075 ± 0.00518	720,000

Table 1 shows strong concentration in the structured Friedman2 circuit: only 5–12 of 48 parameters capture at least 90% of the loss-gradient energy. Friedman1 is less concentrated but still nonuniform, with 22–29 of 80 parameters capturing at least 80%. Since the two blocks use different target ratios, they represent two operating regimes rather than directly ranking the circuit architectures.

On Friedman2, full PSR reaches a test MAE of 0.3684 and SGIQ with $(m, k) = (10, 7)$ reaches 0.3686 while using $(m + k)/d = 17/48 \approx 35\%$ of the PSR gradient circuits per step.

5.3 Friedman1 comparison

We use a fixed Friedman1 split of 200 training, 40 validation, and 80 test examples, with normalization fitted on the training data, and the circuit of Fig. 2. Both environments train for 30 epochs ($B = 20$, $T = 300$ updates) with Adam, $\beta = (0.9, 0.999)$, and the learning rates above. In the ideal environment, $c = 0.1$ satisfies the condition $c \leq (d - k)^{-1/4} T^{-1/8}$ of Corollary 1(a) for every k , since $80^{-1/4} 300^{-1/8} \approx 0.16$. For each of three seeds, all methods share the data split, initialization, and mini-batch order. Configurations are selected by mean best validation MAE, and test errors are reported as mean $\pm$ sample standard deviation over the three runs. Costs are the gradient circuits of Section 4.3, $2BTd$, $2BTm$, and $2BT(m + k)$, excluding forward, evaluation, and search circuits.

We screen SPSA with $m \in \{10, 20, \dots, 80\}$ and the SGIQ configurations listed in Appendix C. Validation selects SPSA-30 and SGIQ (40, 10) in the ideal environment, and SPSA-60 and SGIQ (40, 20) in the finite-shot environment. Table 2 also includes PSR, ideal SPSA-60 as a higher-cost reference, and finite-shot SPSA-20 as a low-cost reference.

In the ideal environment, SGIQ (40, 10) reduces mean test MAE by 9.1% relative to SPSA-60 using 16.7% fewer gradient circuits. It improves on SPSA-30 at higher cost and uses 37.5% fewer circuits than PSR at higher error. In the finite-shot environment, (40, 20) reduces mean test MAE and MSE by 11.2% and 18.9% relative to equal-budget SPSA-60, respectively, and achieves comparable mean errors to PSR with 25% fewer gradient circuits. These results show that the selected configurations provide empirical cost–accuracy trade-offs in both environments (Appendix C).

6 Conclusion

Efficient quantum gradient estimation is essential for backpropagation in quantum neural networks. Motivated by concentrated loss-gradient energy, we introduced SGIQ to allocate measurement

resources according to coordinate importance. Averaged SPSA provides a low-cost global probe, and PSR replaces selected coordinate estimates while SPSA estimates are retained elsewhere. Experiments show lower gradient-circuit cost than PSR and better predictive accuracy than SPSA. These results support sparsity-informed gradient estimation as a practical approach to reducing the cost of quantum backpropagation.

Acknowledgments and Disclosure of Funding

We sincerely thank Dr. Bicheng Ying from Google for his suggestion and help in this work. This work was supported in part by RIT CHAI Faculty Seed Grant, NVIDIA Academic Grant Program, NIH award R16GM159671, and NSF grants 2537471 and 2112471. The content is solely the responsibility of the authors and does not necessarily represent the official views of the funding agencies.

References

- [1] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii. Quantum circuit learning. *Physical Review A*, 98:032309, 2018.
- [2] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021.
- [3] A. Abbas, R. King, H.-Y. Huang, W. J. Huggins, R. Movassagh, D. Gilboa, and J. R. McClean. On quantum backpropagation, information reuse, and cheating measurement collapse. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [4] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran. Evaluating analytic gradients on quantum hardware. *Physical Review A*, 99:032331, 2019.
- [5] A. F. Izmaylov, R. A. Lang, and T.-C. Yen. Analytic gradients in variational quantum algorithms: Algebraic extensions of the parameter-shift rule to general unitary transformations. *Physical Review A*, 104:062443, 2021.
- [6] J. C. Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- [7] B. Coyle, S. Raj, V. Umathe, E. A. Cherrat, and E. Kashefi. Adaptive directional gradients for parameterised quantum circuits. arXiv:2606.09734, 2026.
- [8] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12:1791, 2021.
- [9] A. V. Uvarov and J. D. Biamonte. On barren plateaus and cost function locality in variational quantum algorithms. *Journal of Physics A: Mathematical and Theoretical*, 54(24):245301, 2021.
- [10] M. Heidari, M. Mozakka, and W. Szpankowski. Hadamard test is sufficient for efficient quantum gradient estimation with Lie algebraic symmetries. In *Advances in Neural Information Processing Systems*, volume 38, 2025.
- [11] D. V. Else, F. Machado, C. Nayak, and N. Y. Yao. An improved Lieb–Robinson bound for many-body Hamiltonians with power-law interactions. *Physical Review A*, 101:022333, 2020.
- [12] Z. Ding, T. Ko, J. Yao, L. Lin, and X. Li. Random coordinate descent: A simple alternative for optimizing parameterized quantum circuits. *Physical Review Research*, 6:033029, 2024.
- [13] M. Periyasamy, A. Plinge, C. Mutschler, D. D. Scherer, and W. Maurer. Guided-SPSA: Simultaneous perturbation stochastic approximation assisted by the parameter shift rule. In *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024.
- [14] J. H. Friedman. Multivariate adaptive regression splines. *The Annals of Statistics*, 19(1):1–67, 1991.
- [15] A. Skolik, J. R. McClean, M. Mohseni, P. van der Smagt, and M. Leib. Layerwise learning for quantum neural networks. *Quantum Machine Intelligence*, 3:5, 2021.

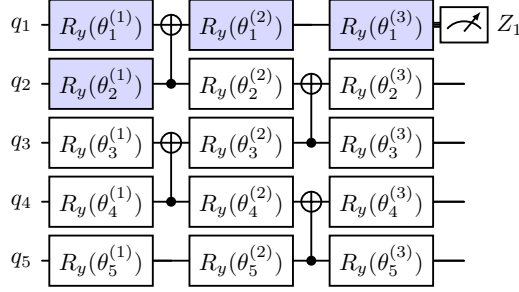


Figure 3: Five-qubit causal-cone example. The shaded rotations are the four parameters in the backward causal cone of Z_1 and are the only rotations that can affect $E_{Z_1}(\theta)$.

A Structural proofs and examples

A.1 Causal-cone proof and five-qubit example

Proof of Proposition 1. For $j \notin \mathcal{C}_O$, backward propagation of each local term O_α through the gates following U_j produces an effective observable $O_{\alpha, > j}$ whose support does not intersect that of G_j . Therefore,

$$[G_j, O_{\alpha, > j}] = 0 \quad \text{for every } \alpha.$$

Using linearity and Eq. 3,

$$\partial_j E_O = \sum_{\alpha} a_{\alpha} \frac{i}{2} \text{Tr}(\rho_j [G_j, O_{\alpha, > j}]) = 0.$$

Thus, $\nabla_{\theta} E_O$ is supported on $\mathcal{C}_O$. If $|\mathcal{C}_O| \leq k$, the support can be extended to a set of size k , and Definition 1 holds with $\rho = 1$. $\square$

Example 1 (Five-qubit local readout). *Consider the five-qubit circuit*

$$U(\theta) = U_R^{(3)} U_E^{(2)} U_R^{(2)} U_E^{(1)} U_R^{(1)}, \quad U_R^{(\ell)} = \prod_{i=1}^5 R_y(\theta_i^{(\ell)}),$$

where

$$U_E^{(1)} = \text{CNOT}_{2 \rightarrow 1} \text{CNOT}_{4 \rightarrow 3}, \quad U_E^{(2)} = \text{CNOT}_{3 \rightarrow 2} \text{CNOT}_{5 \rightarrow 4},$$

and the measured observable is $O = Z_1$.

Propagating Z_1 backward first encounters $R_y(\theta_1^{(3)})$. Neither CNOT in $U_E^{(2)}$ expands its support, so the middle rotation layer contributes only $R_y(\theta_1^{(2)})$. Under $\text{CNOT}_{2 \rightarrow 1}$,

$$Z_1 \mapsto Z_2 Z_1,$$

and hence the first rotation layer contributes $R_y(\theta_1^{(1)})$ and $R_y(\theta_2^{(1)})$.

The backward-cone parameter set is therefore

$$\mathcal{C}_O = \{(1, 1), (1, 2), (2, 1), (3, 1)\},$$

where (ℓ, i) indexes $\theta_i^{(\ell)}$. Only four of the fifteen rotation parameters can have nonzero derivatives. Consequently,

$$\nabla_{\theta} E_{Z_1}$$

is supported on at most four coordinates and is therefore $(4, 1)$ -sparse.

A.2 Commutation proof and diagonal Ising example

Proof of Proposition 2. If $j \notin \mathcal{A}_O(\theta)$, then $[G_j, O_{> j}] = 0$ at this θ , and Eq. 3 gives

$$\partial_j E_O(\theta) = \frac{i}{2} \text{Tr}(\rho_j [G_j, O_{> j}]) = 0.$$

Hence $\nabla_{\theta} E_O(\theta)$ is supported on $\mathcal{A}_O(\theta)$. If $|\mathcal{A}_O(\theta)| \leq k$, the support can be extended to a set of size k , and Definition 1 holds with $\rho = 1$. $\square$

Example 2 (Diagonal Ising block). Let $\mathcal{G} = (V, E)$ be a weighted graph and define

$$O_C = \sum_{(u,v) \in E} w_{uv} Z_u Z_v, \quad U_D(\gamma) = \prod_{(u,v) \in E} e^{-i\gamma_{uv} Z_u Z_v / 2}.$$

Here, V and E are the vertex and edge sets, w_{uv} is the weight of edge (u, v) , and γ_{uv} parameterizes the corresponding diagonal gate. Consider

$$U(\omega, \gamma) = U_D(\gamma) U_M(\omega),$$

where $U_M(\omega)$ is a mixer block with d_M trainable parameters.

Every generator in U_D commutes with every term in O_C . Therefore,

$$U_D^\dagger(\gamma) O_C U_D(\gamma) = O_C$$

for all γ , and

$$\frac{\partial E_{O_C}}{\partial \gamma_{uv}} = 0 \quad \text{for every } (u, v) \in E.$$

The mixer coordinates may remain active; for example,

$$[X_u, Z_u Z_v] \neq 0.$$

The circuit contains

$$d = d_M + |E|$$

parameters but has at most d_M potentially nonzero gradient coordinates. Thus,

$$\nabla E_{O_C}$$

is $(d_M, 1)$ -sparse, and these zeros are structural. For a dense n -vertex graph with $|E| = \Theta(n^2)$ and $d_M = O(n)$, the fraction of potentially active coordinates satisfies

$$\frac{d_M}{d_M + |E|} = O\left(\frac{1}{n}\right).$$

This example demonstrates algebraic sparsity: a parameter can have zero derivative even when its gate is geometrically connected to the measured qubits.

A.3 Proof of the decaying-response bound

Proof of Proposition 3. By linearity, the triangle inequality, and Assumption 1,

$$|\partial_j E_O| \leq \sum_{\alpha} |a_{\alpha}| |\partial_j E_{O_{\alpha}}| \leq L_h \sum_{\alpha} |a_{\alpha}| \chi(\text{dist}(X_{\alpha}, \text{supp } G_j)) = L_h A_j.$$

For $j \notin \mathcal{T}_{\tau}$, $A_j \leq \tau$, so $|\partial_j E_O| \leq L_h \tau$. Summing the squares over the $d - k$ coordinates outside $\mathcal{T}_{\tau}$ gives $t := \|(\nabla E_O)_{\mathcal{T}_{\tau}^c}\|_2^2 \leq (d - k) L_h^2 \tau^2$. Let $a := \|(\nabla E_O)_{\mathcal{T}_{\tau}}\|_2^2$. If $a \geq \frac{\rho}{1-\rho} (d - k) L_h^2 \tau^2$, then $a \geq \frac{\rho}{1-\rho} t$, that is, $a \geq \rho(a + t) = \rho \|\nabla E_O\|_2^2$. This is Eq. 2 with $S = \mathcal{T}_{\tau}$ and $|S| = k$. $\square$

B Proofs of the bias, variance, error, descent, convergence, and cost results

B.1 Estimator, notation, and the common decomposition

Fix $(\theta, \mathcal{B})$ and suppress the conditioning. We write $Q = 1$; for $Q > 1$ every pair (sample, measurement group) is a separate circuit with its own directions and becomes a separate row of Γ , and all statements hold verbatim.

Estimator. For sample i and direction r , let $\Delta^{(i,r)} \in \{-1, +1\}^d$ be Rademacher vectors, independent across i and r . Let $\hat{E}_O$ denote a measured circuit output, with $\hat{E}_O = E_O$ in the ideal setting. SGIQ forms the difference quotient on the circuit output of each sample and chains afterwards,

$$\begin{aligned} \bar{g}_j &= \frac{1}{m} \sum_{r=1}^m \sum_{i \in \mathcal{B}} \lambda_i \frac{\hat{E}_O(\theta + c \Delta^{(i,r)}; \mathbf{x}_i) - \hat{E}_O(\theta - c \Delta^{(i,r)}; \mathbf{x}_i)}{2c \Delta_j^{(i,r)}}, \\ g_j^{\text{PSR}} &= \sum_{i \in \mathcal{B}} \lambda_i \frac{\hat{E}_O(\theta + \frac{\pi}{2} \mathbf{e}_j; \mathbf{x}_i) - \hat{E}_O(\theta - \frac{\pi}{2} \mathbf{e}_j; \mathbf{x}_i)}{2}, \end{aligned} \quad (20)$$

and $\tilde{g}_j = g_j^{\text{PSR}}$ for $j \in \mathcal{S}$, $\tilde{g}_j = \bar{g}_j$ otherwise.

Shot noise. Write $\varepsilon^{(i,r)}$ for the shot error of the SPSSA difference of sample i and direction r , and $\zeta_j := g_j^{\text{PSR}} - g_j$. Both have zero mean, are independent of the directions, and are independent of each other and across (i, r) and j . The PSR circuits are run after $\mathcal{S}$ is chosen, with fresh shots. We set

$$\sigma_S^2 := \frac{1}{4c^2} \sum_{i \in \mathcal{B}} \lambda_i^2 \text{Var}(\varepsilon^{(i,r)}), \quad \sigma_{P,j}^2 := \text{Var}(\zeta_j).$$

A measured output is the mean of N_{sh} independent single-shot outcomes with values in $[-\|O\|_{\text{op}}, \|O\|_{\text{op}}]$, so its variance is at most $\|O\|_{\text{op}}^2/N_{\text{sh}}$. A difference of two independently measured outputs therefore has shot variance at most $2\|O\|_{\text{op}}^2/N_{\text{sh}}$, and ζ_j is $\sum_i \lambda_i/2$ times such differences. With $s = \frac{1}{2}\|O\|_{\text{op}}^2 \sum_i \lambda_i^2$,

$$\sigma_S^2 \leq \frac{1}{4c^2} \sum_i \lambda_i^2 \frac{2\|O\|_{\text{op}}^2}{N_{\text{sh}}} = \frac{s}{N_{\text{sh}}c^2}, \quad \sigma_{P,j}^2 \leq \frac{1}{4} \sum_i \lambda_i^2 \frac{2\|O\|_{\text{op}}^2}{N_{\text{sh}}} = \frac{s}{N_{\text{sh}}},$$

which is Eq. 7. For Pauli observables $\|O\|_{\text{op}} = 1$ and the single-shot variance is $1 - E_O^2$. In the finite-shot setting the factors λ_i are computed from the forward pass, which is independent of the gradient circuits and shared by all three estimators. For the MSE loss with a linear readout, $\partial \ell / \partial \hat{y}_i$ is affine in the measured output, so this estimate $\hat{\lambda}_i$ is unbiased. Hence $\sum_i \hat{\lambda}_i \nabla E_O(\theta; \mathbf{x}_i)$ has mean $\mathbf{g}_B$. We condition on $\hat{\lambda}$ throughout Sections B.1–B.6 and write λ for it. In the finite-shot setting, $\mathbf{g}_{B_t}$ in Assumption 3 denotes this forward-pass gradient $\sum_i \hat{\lambda}_i \nabla E_O(\theta_t; \mathbf{x}_i)$, which has the same mean, so σ_B^2 also bounds the fluctuation of $\hat{\lambda}$.

Regularity. Each θ_j enters a single Pauli rotation, so E_O is a trigonometric polynomial of degree one in every θ_j . Repeated use of the shift rule then gives $|\partial^\alpha E_O| \leq \|O\|_{\text{op}}$ for every multi-index α . All $O(c^p)$ terms below are therefore uniform, with constants depending only on d , $\|O\|_{\text{op}}$ and $\max_i |\lambda_i|$.

Common decomposition. Set $\mathbf{x} = c\Delta^{(i,r)}$. In the Taylor expansion of $E_O(\theta + \mathbf{x}) - E_O(\theta - \mathbf{x})$ the even orders cancel:

$$E_O(\theta + \mathbf{x}) - E_O(\theta - \mathbf{x}) = 2\mathbf{x}^\top \nabla E_O + \frac{1}{3} \sum_{a,b,e} \partial_{abe} E_O x_a x_b x_e + O(c^5).$$

Dividing by $2c\Delta_j^{(i,r)}$ and using $(\Delta_j)^{-1} = \Delta_j$,

$$\hat{u}_j^{(i,r)} = \underbrace{\partial_j E_O + \sum_{p \neq j} \partial_p E_O \Delta_p^{(i,r)} \Delta_j^{(i,r)}}_{=: \xi_{ij}^{(i,r)}} + \underbrace{\frac{c^2}{6} \sum_{a,b,e} \partial_{abe} E_O \Delta_j^{(i,r)} \Delta_a^{(i,r)} \Delta_b^{(i,r)} \Delta_e^{(i,r)}}_{=: r_{ij}^{(i,r)}} + O(c^4). \quad (21)$$

No bound has been used: ξ is the exact coordinate-mixing term and r is the Taylor remainder. Multiplying by λ_i and adding the shot error,

$$\bar{g}_j - g_j = \frac{1}{m} \sum_{r=1}^m \sum_{i \in \mathcal{B}} \lambda_i \left(\xi_{ij}^{(i,r)} + r_{ij}^{(i,r)} + \frac{\varepsilon^{(i,r)} \Delta_j^{(i,r)}}{2c} \right). \quad (22)$$

Fourth moments and the bias coefficient. For Rademacher entries, $\mathbb{E}[\Delta_j \Delta_a \Delta_b \Delta_e] = 1$ if the four indices pair up and 0 otherwise. The three pairings $(a=j, b=e)$, $(b=j, a=e)$ and $(e=j, a=b)$ each contribute $\sum_p \partial_j \partial_p^2 E_O$, and the case $a=b=e=j$ is counted three times instead of once. Hence

$$\mathbb{E}[r_{ij}^{(i,r)}] = c^2 b_{ij} + O(c^4), \quad b_{ij} := \frac{1}{6} \left[3 \sum_p \partial_j \partial_p^2 E_O - 2 \partial_j^3 E_O \right](\theta; \mathbf{x}_i), \quad \beta_{ij} := \lambda_i b_{ij}, \quad (23)$$

so that $\bar{\beta}_j = \sum_i \beta_{ij}$ is the coefficient of Eq. 6.

B.2 Proof of Lemma 1

For $j \in \mathcal{S}$, $\mathbb{E}[g_j^{\text{PSR}}] = g_j$ because the shift rule is exact and ζ_j has zero mean. For $j \notin \mathcal{S}$, take expectations in Eq. 22. For $p \neq j$ the entries Δ_p and Δ_j are independent with zero mean, so $\mathbb{E}[\xi] = 0$. The shot term has zero mean, and Eq. 23 gives the remainder. Under Assumption 2, conditioning on $\mathcal{S}$ does not change these expectations, so $\mathbb{E}[\tilde{g}_j] - g_j = c^2 \sum_i \beta_{ij} + O(c^4) = c^2 \bar{\beta}_j + O(c^4)$.

Squaring and summing over $j \notin \mathcal{S}$ gives $F(k) = \sum_{j \notin \mathcal{S}} (c^2 \bar{\beta}_j + O(c^4))^2 = c^4 \sum_{j \notin \mathcal{S}} \bar{\beta}_j^2 + O((d-k)c^6)$, which is Eq. 8. Exactly, $F(k)$ is the sum over $\mathcal{S}^c$ of the squared coordinate biases of $\bar{\mathbf{g}}$. These biases depend on neither m nor $\mathcal{S}$, so enlarging a nested $\mathcal{S}$ can only remove nonnegative terms, and $F(d) = 0$. $\square$

B.3 Proof of Lemma 2

For $j \in \mathcal{S}$, $\text{Var}(\tilde{g}_j) = \sigma_{\text{P},j}^2$. For $j \notin \mathcal{S}$, the summands of Eq. 22 are independent across (i, r) . The shot term has zero mean and is independent of the directions, so it is uncorrelated with $\xi + r$. Hence

$$\text{Var}(\tilde{g}_j) = \frac{1}{m} \sum_{i \in \mathcal{B}} \lambda_i^2 \left[\text{Var}(\xi_{ij}) + 2 \text{Cov}(\xi_{ij}, r_{ij}) + \text{Var}(r_{ij}) \right] + \frac{\sigma_{\text{S}}^2}{m}.$$

Mixing. Since $\Delta_j^2 = 1$, $\mathbb{E}[\Delta_p \Delta_j \Delta_{p'} \Delta_j] = \delta_{pp'}$, and $\text{Var}(\xi_{ij}) = \sum_{p \neq j} (\partial_p E_O)^2$. *Cross term.* Again using $\Delta_j^2 = 1$ and the fourth-moment count,

$$\text{Cov}(\xi_{ij}, r_{ij}) = \frac{c^2}{6} \sum_{p \neq j} \partial_p E_O \sum_{a,b,e} \partial_{abe} E_O \mathbb{E}[\Delta_p \Delta_a \Delta_b \Delta_e] + O(c^4) = c^2 \sum_{p \neq j} \partial_p E_O b_{ip} + O(c^4).$$

Remainder. $\text{Var}(r_{ij}) = O(c^4)$.

Multiplying by λ_i^2 turns $\partial_p E_O$ and b_{ip} into Γ_{ip} and β_{ip} , and summing over i gives

$$\text{Var}(\tilde{g}_j) = \frac{1}{m} \left[\|\Gamma\|_F^2 - \|\Gamma_{:,j}\|_2^2 \right] + \frac{2c^2}{m} \left[\langle \Gamma, \beta \rangle_F - \sum_i \Gamma_{ij} \beta_{ij} \right] + \frac{\sigma_{\text{S}}^2}{m} + O\left(\frac{c^4}{m}\right).$$

Under Assumption 2, $\text{Var}(\tilde{\mathbf{g}})$ is the sum of these coordinate variances. Summing over $j \notin \mathcal{S}$ with $\sum_{j \notin \mathcal{S}} \|\Gamma_{:,j}\|_2^2 = \|\Gamma_{\mathcal{S}^c}\|_F^2$ and $\sum_{j \notin \mathcal{S}} \sum_i \Gamma_{ij} \beta_{ij} = \langle \Gamma_{\mathcal{S}^c}, \beta_{\mathcal{S}^c} \rangle_F$, and adding $\sum_{j \in \mathcal{S}} \sigma_{\text{P},j}^2$, gives Eq. 9, whose $O((d-k)c^2/m)$ term is $\mathcal{C}_c := \frac{2c^2}{m} [\langle \Gamma, \beta \rangle_F - \langle \Gamma_{\mathcal{S}^c}, \beta_{\mathcal{S}^c} \rangle_F] + O\left(\frac{(d-k)c^4}{m}\right)$. Finally, $0 \leq \|\Gamma_{\mathcal{S}^c}\|_F^2 \leq \|\Gamma\|_F^2$ places the mixing term between $(d-k-1)\|\Gamma\|_F^2/m$ and $(d-k)\|\Gamma\|_F^2/m$ in the ideal setting. $\square$

B.4 Proof of Proposition 4

(i) On $\mathcal{S}$, $\tilde{g}_j - g_j = \zeta_j$; off $\mathcal{S}$, $\tilde{g}_j - g_j = \bar{g}_j - g_j$. Hence, pathwise and for any rule,

$$\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 = \|\bar{\mathbf{g}} - \mathbf{g}\|_2^2 - \sum_{j \in \mathcal{S}} (\bar{g}_j - g_j)^2 + \sum_{j \in \mathcal{S}} \zeta_j^2.$$

The PSR shots are fresh, so $\mathbb{E}[\zeta_j^2 \mid \mathcal{S}, \bar{\mathbf{g}}] = \sigma_{\text{P},j}^2$. Taking expectations gives Eq. 10. In the ideal setting $\zeta_j = 0$ and the subtracted term is nonnegative. Under Assumption 2, the proof of Lemma 2 gives $\mathbb{E}(\bar{g}_j - g_j)^2 \geq \text{Var}(\bar{g}_j) \geq \sigma_{\text{S}}^2/m$, so each term of the subtracted sum is nonnegative when $\sigma_{\text{P}}^2 \leq \sigma_{\text{S}}^2/m$.

(ii) This is the bias-variance identity $\mathbb{E}\|\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 = \|\mathbb{E}\tilde{\mathbf{g}} - \mathbf{g}\|_2^2 + \text{Var}(\tilde{\mathbf{g}})$ combined with Lemmas 1–2. At $k = 0$, $\|\Gamma_{\mathcal{S}^c}\|_F^2 = \|\Gamma\|_F^2$, and the mixing term is $(d-1)\|\Gamma\|_F^2/m$. Hence, in the ideal setting, the mixing term of SGIQ lies between $(d-k-1)\|\Gamma\|_F^2/m$ and $(d-k)\|\Gamma\|_F^2/m$ (proof of Lemma 2), versus $(d-1)\|\Gamma\|_F^2/m$ for averaged SPSA. $\square$

B.5 Proof of Proposition 5

Write $D := \langle \mathbf{g}, \tilde{\mathbf{g}} - \mathbf{g} \rangle = \sum_{j \notin \mathcal{S}} g_j (\bar{g}_j - g_j) + \sum_{j \in \mathcal{S}} g_j \zeta_j$. Since $\langle \mathbf{g}, \mathbf{g} \rangle$ is deterministic, $\text{Var}(\langle \mathbf{g}, \tilde{\mathbf{g}} \rangle) = \text{Var}(D)$. The mean follows from Lemma 1. By Eq. 22, $D = M + R + N_{\mathcal{S}} + N_{\mathcal{P}}$

with mixing part M , remainder part R , SPSA shot part N_S and PSR shot part N_P . The shot parts have zero mean, are independent of each other, and are independent of the directions, so they are uncorrelated with $M + R$. Moreover $\text{Var}(R) = O(c^4/m)$ and $\text{Cov}(M, R) = O(c^2/m)$.

Mixing part. $M = \frac{1}{m} \sum_r \sum_i X_i^{(r)}$ with $X_i = \sum_{j \notin \mathcal{S}} \sum_{p \neq j} g_j \Gamma_{ip} \Delta_j \Delta_p$. The summands are independent across (i, r) , so $\text{Var}(M) = m^{-1} \sum_i V_i$ with $V_i := \text{Var}(X_i)$. Group the terms of X_i by unordered pairs $\{j, p\}$. A pair with both indices in $\mathcal{S}^c$ appears twice, with coefficient $g_j \Gamma_{ip} + g_p \Gamma_{ij}$. A pair with $j \notin \mathcal{S}$ and $p \in \mathcal{S}$ appears once, with coefficient $g_j \Gamma_{ip}$. A pair inside $\mathcal{S}$ does not appear. The products $\Delta_j \Delta_p$ over distinct pairs have zero mean, unit variance, and are pairwise uncorrelated, so

$$V_i = \sum_{\{j,p\} \subseteq \mathcal{S}^c} (g_j \Gamma_{ip} + g_p \Gamma_{ij})^2 + \sum_{j \notin \mathcal{S}, p \in \mathcal{S}} g_j^2 \Gamma_{ip}^2.$$

By $(a+b)^2 \leq 2a^2 + 2b^2$, the first sum is at most $2\|g_{\mathcal{S}^c}\|_2^2 \|\Gamma_{i, \mathcal{S}^c}\|_2^2$. The second equals $\|g_{\mathcal{S}^c}\|_2^2 \|\Gamma_{i, \mathcal{S}}\|_2^2$. Hence $V_i \leq 2\|g_{\mathcal{S}^c}\|_2^2 \|\Gamma_i\|_2^2$, and, with $\|g_{\mathcal{S}^c}\|_2^2 = (1 - \hat{\rho})\|g\|_2^2$, $\text{Var}(M) \leq 2(1 - \hat{\rho})\|g\|_2^2 \|\Gamma\|_F^2/m$.

SPSA shot part. $N_S = \frac{1}{m} \sum_{r,i} \lambda_i \frac{\varepsilon^{(i,r)}}{2c} \sum_{j \notin \mathcal{S}} g_j \Delta_j^{(i,r)}$. Since $\mathbb{E}(\sum_{j \notin \mathcal{S}} g_j \Delta_j)^2 = \|g_{\mathcal{S}^c}\|_2^2$, $\text{Var}(N_S) = (1 - \hat{\rho})\|g\|_2^2 \sigma_S^2/m$.

PSR shot part. $\text{Var}(N_P) = \sum_{j \in \mathcal{S}} g_j^2 \sigma_{P,j}^2 \leq \hat{\rho} \|g\|_2^2 \sigma_P^2$.

Adding the four parts proves Eq. 11. $\square$

B.6 Finite shots: the perturbation radius and the difference quotient

Choice of c . By Proposition 4(ii), the c -dependent part of the error is $f(c) = \kappa_{\text{cr}} c^2 + \kappa_{\text{bs}} c^4 + \kappa_{\text{sh}} c^{-2}$, with

$$\kappa_{\text{cr}} = \frac{2}{m} [(d-k) \langle \Gamma, \beta \rangle_F - \langle \Gamma_{\mathcal{S}^c}, \beta_{\mathcal{S}^c} \rangle_F], \quad \kappa_{\text{bs}} = \|\bar{\beta}_{\mathcal{S}^c}\|_2^2 \geq 0, \\ \kappa_{\text{sh}} = \frac{(d-k)c^2 \sigma_S^2}{m} = \Theta\left(\frac{d-k}{m N_{\text{sh}}}\right).$$

If $\kappa_{\text{bs}} > 0$, then $f \rightarrow \infty$ at both ends, so a finite minimizer exists, and it satisfies $2\kappa_{\text{bs}} c^6 + \kappa_{\text{cr}} c^4 = \kappa_{\text{sh}}$. When the cross term is negligible, $c_\star = (\kappa_{\text{sh}}/2\kappa_{\text{bs}})^{1/6} = \Theta(N_{\text{sh}}^{-1/6})$. If $\kappa_{\text{cr}} > 0$ dominates at small c , then $c_\star = \Theta(N_{\text{sh}}^{-1/4})$. If $\kappa_{\text{cr}} < 0$, then $c_\star \rightarrow \sqrt{|\kappa_{\text{cr}}|/(2\kappa_{\text{bs}})}$ as $N_{\text{sh}} \rightarrow \infty$. The sign of κ_{cr} depends on the circuit, so we treat c as a tuned hyperparameter.

Why the quotient is formed on circuit outputs. Suppose instead the quotient were formed on a shot estimate $\hat{\mathcal{L}}_{\mathcal{B}}$ of the scalar loss. Write $\hat{g}_i = y_i^\star + \delta_i$ with $\mathbb{E}[\delta_i] = 0$ and $\text{Var}(\delta_i) = \nu_i/N_{\text{sh}}$. For the MSE loss this gives the exact identity

$$\mathbb{E}[\hat{\mathcal{L}}_{\mathcal{B}}(\theta)] = \mathcal{L}_{\mathcal{B}}(\theta) + \varsigma^2(\theta), \quad \varsigma^2(\theta) := \frac{1}{|\mathcal{B}| N_{\text{sh}}} \sum_{i \in \mathcal{B}} \nu_i(\theta). \quad (24)$$

This holds at every evaluation point, so such a scheme performs SPSA on $\mathcal{L}_{\mathcal{B}} + \varsigma^2$. Its bias contains $\partial_j \varsigma^2 = O(1/N_{\text{sh}})$, which depends on neither c nor m , and it pushes the optimizer towards states of small readout variance. Eq. 20 uses the same circuit evaluations but never squares a measured quantity before differencing, so its bias remains that of Lemma 1.

B.7 Proof of Theorem 1

Let $\mathbb{E}_t$ condition on the history before step t , and let $\mathbf{G}_t := \nabla \mathcal{J}(\theta_t)$, $\mathbf{b}_t := \mathbb{E}_t[\tilde{\mathbf{g}}_t] - \mathbf{G}_t$ and $v_t := \text{Var}_t(\tilde{\mathbf{g}}_t)$.

Step 1: descent inequality for any estimator. Assume $\|\mathbf{b}_t\|_2^2 \leq b^2$ and $v_t \leq v$ for all t . By L_g -smoothness, $\mathcal{J}(\theta_{t+1}) \leq \mathcal{J}(\theta_t) - \eta \langle \mathbf{G}_t, \tilde{\mathbf{g}}_t \rangle + \frac{L_g \eta^2}{2} \|\tilde{\mathbf{g}}_t\|_2^2$. Here $\mathbb{E}_t \langle \mathbf{G}_t, \tilde{\mathbf{g}}_t \rangle = \|\mathbf{G}_t\|_2^2 + \langle \mathbf{G}_t, \mathbf{b}_t \rangle$ and $\mathbb{E}_t \|\tilde{\mathbf{g}}_t\|_2^2 = \|\mathbf{G}_t + \mathbf{b}_t\|_2^2 + v_t \leq 2\|\mathbf{G}_t\|_2^2 + 2b^2 + v$. Young's inequality gives $-\langle \mathbf{G}_t, \mathbf{b}_t \rangle \leq \frac{1}{4} \|\mathbf{G}_t\|_2^2 + b^2$. With $L_g \eta \leq \frac{1}{4}$,

$$\mathbb{E}_t \mathcal{J}(\theta_{t+1}) \leq \mathcal{J}(\theta_t) - \frac{\eta}{2} \|\mathbf{G}_t\|_2^2 + \frac{5\eta}{4} b^2 + \frac{L_g \eta^2}{2} v.$$

Taking total expectations, summing over $t = 0, \dots, T-1$, using $\mathcal{J}(\theta_T) \geq \mathcal{J}_\star$ and dividing by $\eta T/2$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\mathbf{G}_t\|_2^2 \leq \frac{2\Delta_{\mathcal{J}}}{\eta T} + L_g \eta v + \frac{5}{2} b^2 \leq \frac{2\Delta_{\mathcal{J}}}{\eta T} + L_g \eta v + 3b^2. \quad (25)$$

Step 2: part (i). Condition on $\mathcal{B}_t$. By Lemma 1, $\mathbb{E}[\tilde{\mathbf{g}}_t | \mathcal{B}_t] = \mathbf{g}_{\mathcal{B}_t} + \boldsymbol{\beta}_t$ with $\|\boldsymbol{\beta}_t\|_2^2 \leq F(k)$. Since $\mathbb{E}_t \mathbf{g}_{\mathcal{B}_t} = \mathbf{G}_t$, Jensen's inequality gives $\|\mathbf{b}_t\|_2^2 = \|\mathbb{E}_t \tilde{\mathbf{g}}_t\|_2^2 \leq F(k)$. By the law of total variance,

$$v_t = \mathbb{E}_t [\text{Var}(\tilde{\mathbf{g}}_t | \mathcal{B}_t)] + \text{Var}_t(\mathbf{g}_{\mathcal{B}_t} + \boldsymbol{\beta}_t).$$

The first term is at most $V(k)$ by Lemma 2, after dropping $-\|\Gamma_{\mathcal{S}^c}\|_F^2/m$ and using $\|\Gamma_t\|_F^2 \leq \bar{G}$, $\sigma_{\mathcal{P},j}^2 \leq \sigma_{\mathcal{P}}^2 \leq \bar{s}/N_{\text{sh}}$, $\sigma_{\mathcal{S}}^2 \leq \bar{s}/(N_{\text{sh}} c^2)$ (Eq. 7) and $|\mathcal{C}_c| = O((d-k)c^2/m)$; likewise $\|\boldsymbol{\beta}_t\|_2^2 \leq (d-k)c^4 B_{\beta}^2 + O((d-k)c^6)$ by Lemma 1 and the definition of B_{β}^2 . For the second term, Minkowski's inequality for random vectors gives $\sqrt{\text{Var}(X+Y)} \leq \sqrt{\text{Var}(X)} + \sqrt{\text{Var}(Y)}$. Here $\text{Var}_t(\mathbf{g}_{\mathcal{B}_t}) \leq \sigma_{\mathcal{B}}^2$ and $\text{Var}_t(\boldsymbol{\beta}_t) \leq \mathbb{E}_t \|\boldsymbol{\beta}_t\|_2^2 \leq F(k)$. Hence $v_t \leq V(k) + (\sigma_{\mathcal{B}} + \sqrt{F(k)})^2 = v(k)$, and Eq. 25 gives Eq. 14.

Step 3: part (ii). No property of $\mathcal{S}_t$ is used. Since $\mathbb{E}_t \mathbf{g}_{\mathcal{B}_t} = \mathbf{G}_t$, $\mathbf{b}_t = \mathbb{E}_t[\tilde{\mathbf{g}}_t - \mathbf{g}_{\mathcal{B}_t}]$, and Jensen's inequality gives $\|\mathbf{b}_t\|_2^2 \leq \mathbb{E}_t \|\tilde{\mathbf{g}}_t - \mathbf{g}_{\mathcal{B}_t}\|_2^2 \leq e^2$. By Minkowski's inequality in L^2 , $v_t \leq \mathbb{E}_t \|\tilde{\mathbf{g}}_t - \mathbf{G}_t\|_2^2 \leq (e + \sigma_{\mathcal{B}})^2$. Eq. 25 completes the proof. $\square$

B.8 Proofs of Corollaries 1–3

Corollary 1. Since $T \geq 1$, $\eta = 1/(4L_g \sqrt{T}) \leq 1/(4L_g)$ and Theorem 1 applies, with $2\Delta_{\mathcal{J}}/(\eta T) = 8L_g \Delta_{\mathcal{J}}/\sqrt{T}$ and $L_g \eta v(k) = v(k)/(4\sqrt{T})$. By $(a+b)^2 \leq 2a^2 + 2b^2$, $v(k) \leq V(k) + 2\sigma_{\mathcal{B}}^2 + 2F(k)$, so the bound of part (i) is at most

$$\frac{1}{\sqrt{T}} \left[8L_g \Delta_{\mathcal{J}} + \frac{\sigma_{\mathcal{B}}^2}{2} + \frac{V(k)}{4} \right] + \left(3 + \frac{1}{2\sqrt{T}} \right) F(k), \quad 3 + \frac{1}{2\sqrt{T}} \leq \frac{7}{2}.$$

Inserting $V(k)$ from Eq. 15 and $F(k) \leq (d-k)c^4 B_{\beta}^2 + O((d-k)c^6)$ gives Eq. 16; the $O((d-k)c^2/m)$ term of $V(k)$ contributes $O((d-k)c^2/(m\sqrt{T}))$.

(a) With $\bar{s} = 0$ and $c \leq (d-k)^{-1/4} T^{-1/8}$, we have $(d-k)c^4 \leq T^{-1/2}$, so the floor is at most $\frac{7}{2} B_{\beta}^2/\sqrt{T}$. The neglected terms satisfy $(d-k)c^2/(m\sqrt{T}) \leq (d-k)^{1/2} T^{-3/4}/m$ and $(d-k)c^6 \leq (d-k)^{-1/2} T^{-3/4}$, both $O(T^{-3/4})$. This gives Eq. 17. For $k = d$ every coordinate is estimated by PSR, $F = V = 0$, and the bound is $A_0/\sqrt{T}$ for every c .

(b) For fixed c the bracket of Eq. 16 is a constant, which gives the first claim. The two terms that depend on c are $a c^{-2} + b c^4$ with $a = (d-k)\bar{s}/(4mN_{\text{sh}}\sqrt{T})$ and $b = \frac{7}{2}(d-k)B_{\beta}^2$. Setting the derivative $-2ac^{-3} + 4bc^3$ to zero gives $c_{\star}^6 = a/(2b) = \bar{s}/(28mN_{\text{sh}}B_{\beta}^2\sqrt{T})$, and the minimum value is

$$\frac{3}{2} (2b)^{1/3} a^{2/3} = \frac{3}{2} \cdot 7^{1/3} 4^{-2/3} (d-k) \left(\frac{B_{\beta} \bar{s}}{mN_{\text{sh}}} \right)^{2/3} T^{-1/3} \leq C_{\text{sh}} T^{-1/3},$$

since $\frac{3}{2} \cdot 7^{1/3} 4^{-2/3} \approx 1.14 < 1.2$. The remaining terms of the bracket give $K_{\text{sh}}/\sqrt{T}$. At $c = c_{\star}$ the neglected terms are $O((d-k)T^{-1/2})$ and $O(T^{-2/3})$, of lower order than $T^{-1/3}$.

For part (ii) of Theorem 1, the same substitution gives $\frac{1}{\sqrt{T}} [8L_g \Delta_{\mathcal{J}} + \frac{1}{4}(e + \sigma_{\mathcal{B}})^2] + 3e^2$. $\square$

Corollary 2. (i) The coordinate biases of $\bar{\mathbf{g}}$ do not depend on m (Lemma 1). Hence $F_{\text{SPSA}} - F_{\text{SGiQ}}$ is the sum over $\mathcal{S}$ of these squared biases, which is nonnegative and equals $c^4 \|\bar{\boldsymbol{\beta}}_{\mathcal{S}}\|_2^2 + O(c^6)$. (ii) Proposition 5 with $\hat{\rho} = 0$ and m' directions gives the SPSA bound $2\|\mathbf{g}\|_2^2 \|\Gamma\|_F^2/m'$, and $2(1-\hat{\rho})/m < 2/(m+k)$ if and only if $\hat{\rho} > k/(m+k)$. $\square$

Corollary 3. Each step evaluates, per sample and measurement group, two circuits per PSR coordinate and two per SPSA direction (Section 4.3); multiplying by T and by N_{sh} gives the totals.

In the ideal setting, Eq. 17 is at most ε , up to its $O(T^{-3/4})$ term, once $K(m, k, d)/\sqrt{T} \leq \varepsilon$, that is, $T \geq K(m, k, d)^2/\varepsilon^2$. Multiplying by the per-step cost $2BQ(m+k)$ gives $C_{\text{SGIQ}}(\varepsilon)$. For PSR ($k = d$), $K = A_0$ and the per-step cost is $2BQd$, which gives $C_{\text{PSR}}(\varepsilon)$. Under finite shots with $c = c_*$, $T \geq (2K_{\text{sh}}/\varepsilon)^2$ gives $K_{\text{sh}}/\sqrt{T} \leq \varepsilon/2$, and $T \geq (2C_{\text{sh}}/\varepsilon)^3$ gives $C_{\text{sh}}T^{-1/3} \leq \varepsilon/2$. Hence the bound is at most ε . $\square$

B.9 Proof of Lemma 3

Fix θ and $\mathcal{B}$, and write $e := \bar{g} - g$ and $\epsilon := \max_j |e_j|$.

Relation between $\hat{\rho}$ and ρ_k . Since $\mathcal{S}_*$ maximizes $\|g_S\|_2$ over $|S| = k$, $\hat{\rho} \leq \rho_k$. Since $\mathcal{S}$ maximizes $\|\bar{g}_S\|_2$ over $|S| = k$, and $\|e_S\|_2 \leq \sqrt{k}\epsilon$ for every $|S| = k$,

$$\|g_S\|_2 \geq \|\bar{g}_S\|_2 - \sqrt{k}\epsilon \geq \|\bar{g}_{\mathcal{S}_*}\|_2 - \sqrt{k}\epsilon \geq \|g_{\mathcal{S}_*}\|_2 - 2\sqrt{k}\epsilon.$$

Dividing by $\|g\|_2$ gives Eq. 12.

Deterministic recovery. If $\epsilon < \gamma/2$, then for $j \in \mathcal{S}_*$ and $l \notin \mathcal{S}_*$ we have $|g_j| \geq |g_l| + \gamma$ and

$$|\bar{g}_j| > |g_j| - \frac{\gamma}{2} \geq |g_l| + \frac{\gamma}{2} > |\bar{g}_l|,$$

so every score in $\mathcal{S}_*$ strictly exceeds every score outside it, and $\text{TopK}(|\bar{g}|, k) = \mathcal{S}_*$.

Probability of recovery. Write $\bar{g}_j = \frac{1}{m} \sum_{r=1}^m X_j^{(r)}$, where $X_j^{(r)} := \sum_{i \in \mathcal{B}} \lambda_i (\hat{u}_{ij}^{(r)} + \varepsilon^{(i,r)} \Delta_j^{(i,r)} / (2c))$ collects the contribution of the r -th direction over the batch. The $X_j^{(r)}$ are i.i.d. in r , with mean μ_j satisfying $|\mu_j - g_j| \leq b_\infty$ (Lemma 1) and variance at most $\sigma_X^2 := \|\Gamma\|_F^2 + \sigma_S^2 + O(c^2)$ (the single-direction case of Lemma 2). They are bounded: measured outputs satisfy $|\hat{E}_O| \leq \|O\|_{\text{op}}$, so each difference quotient is at most $\|O\|_{\text{op}}/c$ in magnitude, and in the ideal setting also at most $\|\nabla E_O\|_1 \leq d\|O\|_{\text{op}}$ by the mean-value theorem. Hence $|X_j^{(r)} - \mu_j| \leq R := 2\Lambda\|O\|_{\text{op}}r_c$ with $\Lambda := \sum_i |\lambda_i|$, where $r_c = \min\{d, 1/c\}$ in the ideal setting and $r_c = 1/c$ under finite shots. Bernstein's inequality gives, for every j ,

$$\Pr(|\bar{g}_j - \mu_j| \geq t_\gamma) \leq 2 \exp\left(-\frac{m t_\gamma^2}{2\sigma_X^2 + \frac{2}{3} R t_\gamma}\right).$$

Since $|e_j| \leq |\bar{g}_j - \mu_j| + b_\infty$, the event $\epsilon \geq \gamma/2$ implies $|\bar{g}_j - \mu_j| \geq t_\gamma$ for some j , and a union bound over the d coordinates gives Eq. 13.

Consequences. On the recovery event, $\hat{\rho} = \rho_k$, and $\bar{g}$ coincides with the estimator that uses the fixed set $\mathcal{S}_*$ with the same directions and shots; this estimator satisfies Assumption 2. Since $\hat{\rho} \geq 0$, $\mathbb{E}[\hat{\rho}] \geq \rho_k \Pr(\mathcal{S} = \mathcal{S}_*) \geq (1 - p_{\text{fail}})\rho_k$. Along a run, if the gap holds at every step, an induction over t shows that the iterates of Algorithm 1 and of the predictable variant using $\mathcal{S}_{*,t}$ coincide on the event A that every step recovers $\mathcal{S}_{*,t}$, and a union bound over the T steps gives $\Pr(A^c) \leq T p_{\text{fail}}$. Let $X := \frac{1}{T} \sum_{t < T} \|\nabla \mathcal{J}(\theta_t)\|_2^2 \in [0, G_\infty]$ for Algorithm 1 and X_* the same quantity for the variant. Since $X = X_*$ on A , $\mathbb{E}[X] \leq \mathbb{E}[X_*] + G_\infty \Pr(A^c) \leq \mathbb{E}[X_*] + T p_{\text{fail}} G_\infty$. By Eq. 13, $T p_{\text{fail}} G_\infty \leq \varepsilon$ once $m \geq (2\sigma_X^2 + \frac{2}{3} R t_\gamma) t_\gamma^{-2} \log(2dT G_\infty/\varepsilon)$. $\square$

C Parameter screening and additional results

We evaluate SPSA with $m = 10, 20, \dots, 80$, covering gradient budgets from 12.5% to 100% of PSR. The ideal SGiQ candidates are (20, 10), (22, 8), (30, 10), (40, 10); finite-shot candidates are (10, 10), (12, 8), (20, 10), (30, 10), (40, 10), (40, 20). The latter span 25%–75% of the PSR budget. All configurations use the same three seeds and data split.

Within each run, minimum validation MAE selects the checkpoint; configurations are ranked by the mean of the three run-specific minima. These minima may occur at different epochs. The reported test scores use these checkpoints without further training. The tables below report the full screening together with the available test errors. SPSA-60 is the validation-selected finite-shot baseline; SPSA-50 has the lowest SPSA test MAE (0.11148). The experiments are exploratory: test results were inspected during development, so these are not independent confirmation results.

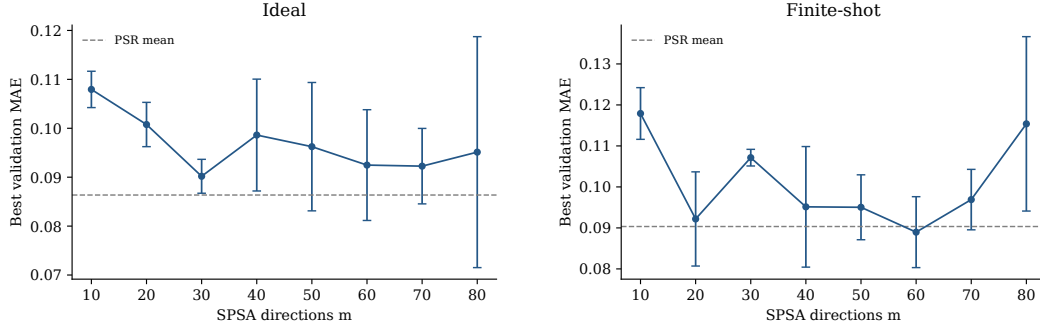


Figure 4: SPSA validation screening. Points and error bars show means and sample standard deviations across three seeds. PSR means are shown as references. Lines connect evaluated configurations only.

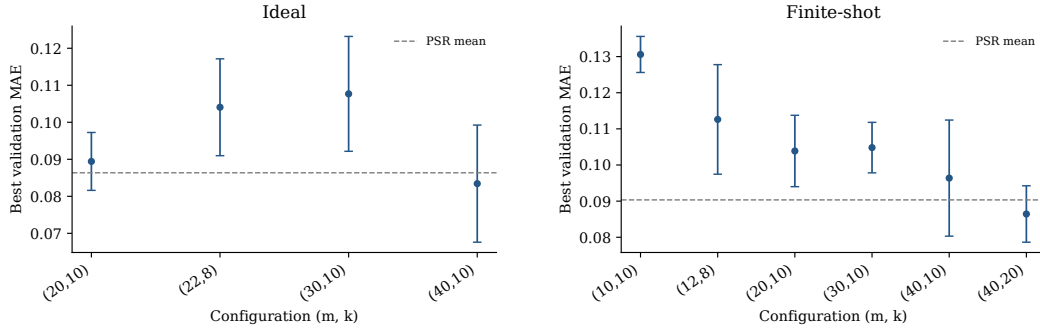


Figure 5: SGiQ validation screening. Labels indicate (m, k) . At fixed $k = 10$, the ideal results are non-monotonic in m . In the finite-shot experiment, increasing k from 10 to 20 at $m = 40$ reduces mean best validation MAE from 0.09637 to 0.08644, while increasing the gradient budget by 20%.

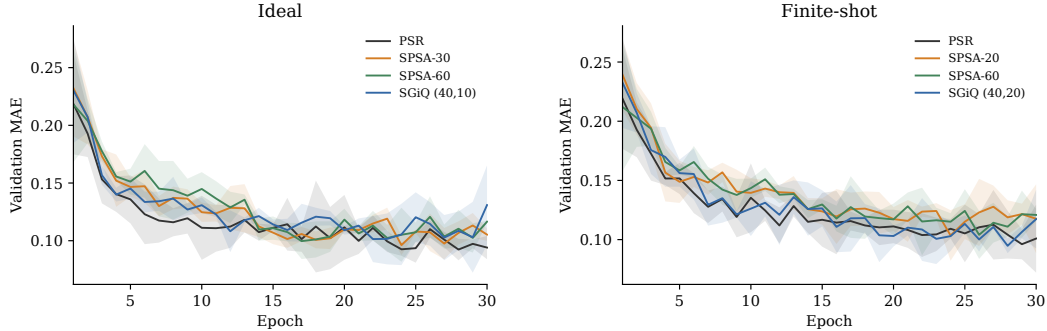


Figure 6: Per-epoch validation MAE of the main-text configurations. Curves show means and shaded bands show sample standard deviations across three seeds. These are end-of-epoch scores, not running minima. Test errors are evaluated only at validation-selected checkpoints.

Table 3: Ideal screening and available tests under Adam (0.9, 0.999). Values are means $\pm$ sample standard deviations over three seeds. A dash denotes an unavailable test summary.

Method	Best val. MAE	Test MAE	Test MSE	Cost
PSR	0.08635 $\pm$ 0.00807	0.08489 $\pm$ 0.01192	0.01304 $\pm$ 0.00261	960,000
SPSA-10	0.10794 $\pm$ 0.00371	0.12177 $\pm$ 0.01907	0.02688 $\pm$ 0.00813	120,000
SPSA-20	0.10077 $\pm$ 0.00453	0.10984 $\pm$ 0.00880	0.02151 $\pm$ 0.00341	240,000
SPSA-30	0.09019 $\pm$ 0.00347	0.10924 $\pm$ 0.01759	0.02122 $\pm$ 0.00370	360,000
SPSA-40	0.09861 $\pm$ 0.01143	0.10496 $\pm$ 0.00338	0.01991 $\pm$ 0.00308	480,000
SPSA-50	0.09624 $\pm$ 0.01312	0.10456 $\pm$ 0.01092	0.02087 $\pm$ 0.00504	600,000
SPSA-60	0.09247 $\pm$ 0.01132	0.10454 $\pm$ 0.01799	0.01888 $\pm$ 0.00601	720,000
SPSA-70	0.09226 $\pm$ 0.00771	0.10488 $\pm$ 0.00999	0.02033 $\pm$ 0.00490	840,000
SPSA-80	0.09512 $\pm$ 0.02361	0.09418 $\pm$ 0.01344	0.01741 $\pm$ 0.00351	960,000
SGiQ (20,10)	0.08943 $\pm$ 0.00782	0.11976 $\pm$ 0.01136	0.02519 $\pm$ 0.00723	360,000
SGiQ (22,8)	0.10406 $\pm$ 0.01308	–	–	360,000
SGiQ (30,10)	0.10768 $\pm$ 0.01552	–	–	480,000
SGiQ (40,10)	0.08343 $\pm$ 0.01582	0.09504 $\pm$ 0.00342	0.01647 $\pm$ 0.00197	600,000

Table 4: Finite-shot screening and available tests under Adam (0.9, 0.999). Values are means $\pm$ sample standard deviations over three seeds. A dash denotes an unavailable test summary.

Method	Best val. MAE	Test MAE	Test MSE	Cost
PSR	0.09034 $\pm$ 0.01967	0.10865 $\pm$ 0.02196	0.02094 $\pm$ 0.01101	960,000
SPSA-10	0.11791 $\pm$ 0.00631	0.14602 $\pm$ 0.00621	0.03733 $\pm$ 0.00558	120,000
SPSA-20	0.09219 $\pm$ 0.01149	0.11579 $\pm$ 0.01090	0.02265 $\pm$ 0.00449	240,000
SPSA-30	0.10714 $\pm$ 0.00203	0.12727 $\pm$ 0.00227	0.02630 $\pm$ 0.00248	360,000
SPSA-40	0.09514 $\pm$ 0.01471	0.11657 $\pm$ 0.00938	0.02339 $\pm$ 0.00600	480,000
SPSA-50	0.09503 $\pm$ 0.00792	0.11148 $\pm$ 0.01654	0.02288 $\pm$ 0.00643	600,000
SPSA-60	0.08897 $\pm$ 0.00865	0.12070 $\pm$ 0.00264	0.02559 $\pm$ 0.00317	720,000
SPSA-70	0.09691 $\pm$ 0.00737	0.11249 $\pm$ 0.00733	0.02076 $\pm$ 0.00104	840,000
SPSA-80	0.11538 $\pm$ 0.02127	0.12000 $\pm$ 0.01074	0.02541 $\pm$ 0.00090	960,000
SGiQ (10,10)	0.13059 $\pm$ 0.00499	–	–	240,000
SGiQ (12,8)	0.11261 $\pm$ 0.01516	–	–	240,000
SGiQ (20,10)	0.10388 $\pm$ 0.00988	–	–	360,000
SGiQ (30,10)	0.10481 $\pm$ 0.00699	–	–	480,000
SGiQ (40,10)	0.09637 $\pm$ 0.01606	–	–	600,000
SGiQ (40,20)	0.08644 $\pm$ 0.00781	0.10714 $\pm$ 0.01119	0.02075 $\pm$ 0.00518	720,000

Budget comparisons. Under ideal simulation, (40, 10) also improves mean test MAE over equal-budget SPSA-50 (0.09504 versus 0.10456), whereas (20, 10) does not improve over equal-budget SPSA-30 (0.11976 versus 0.10924). Under finite shots, (40, 20) improves mean test MAE over SPSA-50 (0.10714 versus 0.11148) at 20% higher cost. Thus the benefit depends on the allocation and environment. The three-run standard deviations describe variability on one fixed split; they do not establish statistical significance or variation across data splits.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The claims are supported by the theory in Sections 3–4 (proofs in Appendices A–B) and the experiments in Section 5.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 4.2 states the assumptions under which each guarantee holds, and Section 5 and Appendix C specify the experimental scope.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: All assumptions are stated with the results in Sections 3–4, and complete proofs are in Appendices A–B.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Algorithm 1, Section 5, and Appendix C specify the estimator, data, circuits, and all training settings, and all methods share the same data split, initialization, and mini-batch order per seed.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The code is not released; the data are synthetic Friedman benchmarks generated as described in Section 5.1.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 5.1 and 5.3 give the data splits, optimizer, and hyperparameters, and Appendix C describes how they were selected.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 2 and Appendix C report the mean $\pm$ sample standard deviation over three seeds.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: The experiments are small classical simulations; their cost is reported as gradient-circuit evaluations (Table 2, Appendix C).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses only synthetic data and simulation and involves no human subjects.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This is foundational work on gradient estimation with no direct societal impact.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no data or models with a risk of misuse.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The Friedman benchmarks are cited [14], and the simulations use Qiskit Aer (Apache-2.0 license).

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.

- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not release new assets.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs were used only for writing assistance and are not part of the method.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.